

Probabilistic Machine Learning

Lecture 03: Linear Models — Keep the Peak, Model Only $p(y | x)$

Prof. Dr. Michael Granitzer

Speaker notes

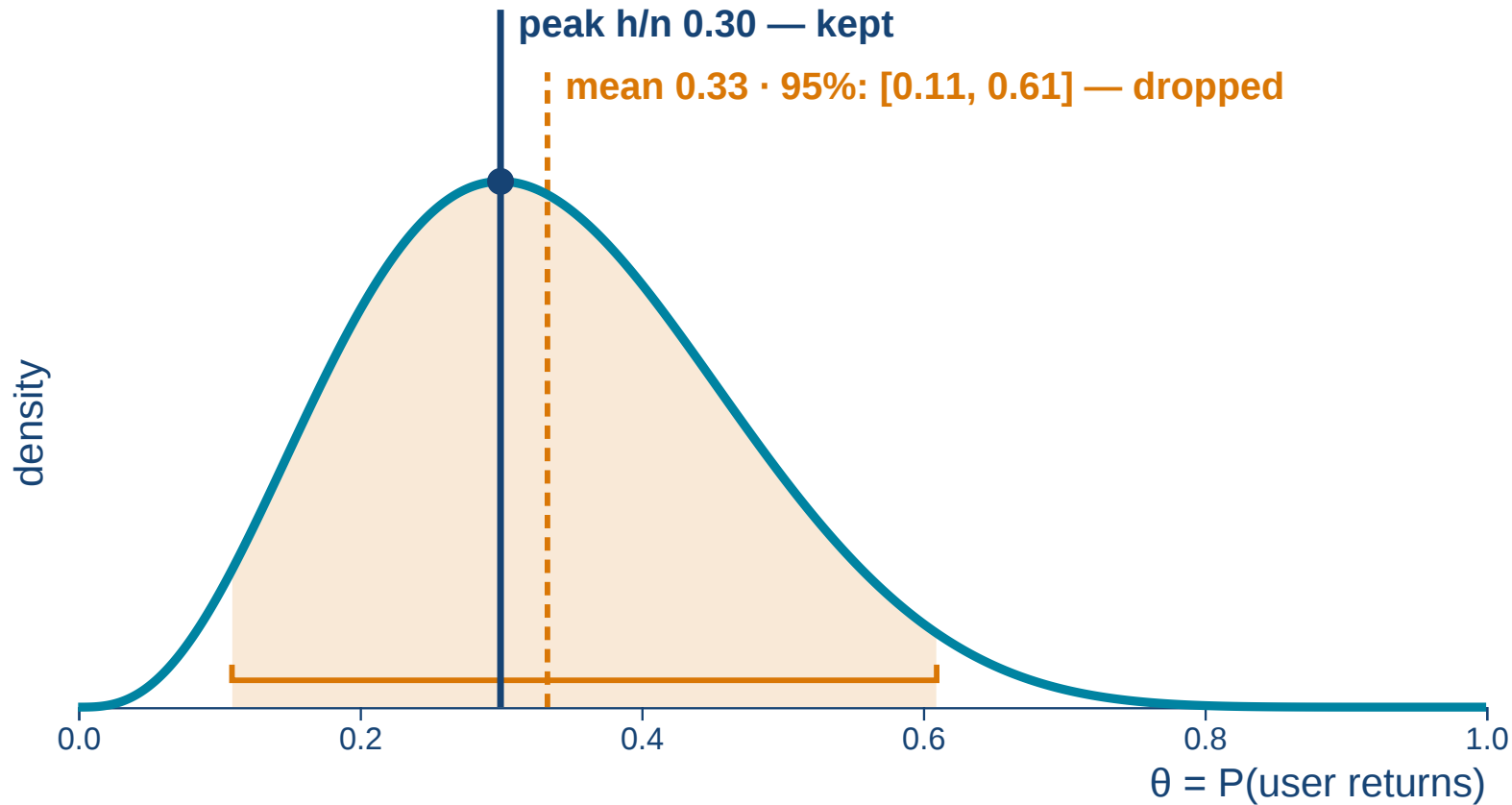
One session. The lecture takes the first two shortcuts of the course on methods the students already know: keep the peak (maximum likelihood and MAP, Modules 1–2) and model only $p(y | x)$ (discriminative, Module 3), and it answers Lecture 01's open question of when a score is a probability (Module 4). Running examples: Lecture 02's returning users (a coin), house prices for flats of 40–120 m² and the 600 m² villa of Lecture 01 (regression), and fraud with a base rate of 1 in 1,000 (classification).

Classical linear regression is one recap slide (Module 2); the closed form, R^2 , basis functions, Gauss–Markov and gradient descent are in the backup after the last module.



Where We Left Off

Lecture 02: 3 of 10 users, flat prior $\rightarrow$ Beta(4, 8)



One posterior, three readings:

- **peak** $0.30 = h/n$
- **mean** 0.33
- **95% interval** [0.11, 0.61]

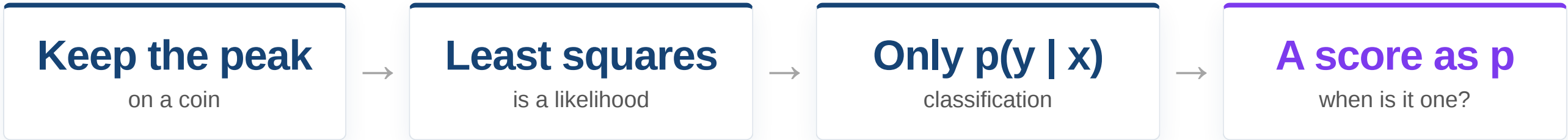
Today keeps only the **peak** — and you already know the methods that do.

Pick up the thread of Lecture 02 deliberately; the last slides there promised this. Show the posterior the students computed last week, fully read: the peak (navy) is what today's methods keep, the mean and the interval (amber) are what they drop. Then say the uncomfortable sentence: every model they have fitted by least squares or cross-entropy took this shortcut. The next slide names the two shortcuts of the lecture.

Today: Two Shortcuts

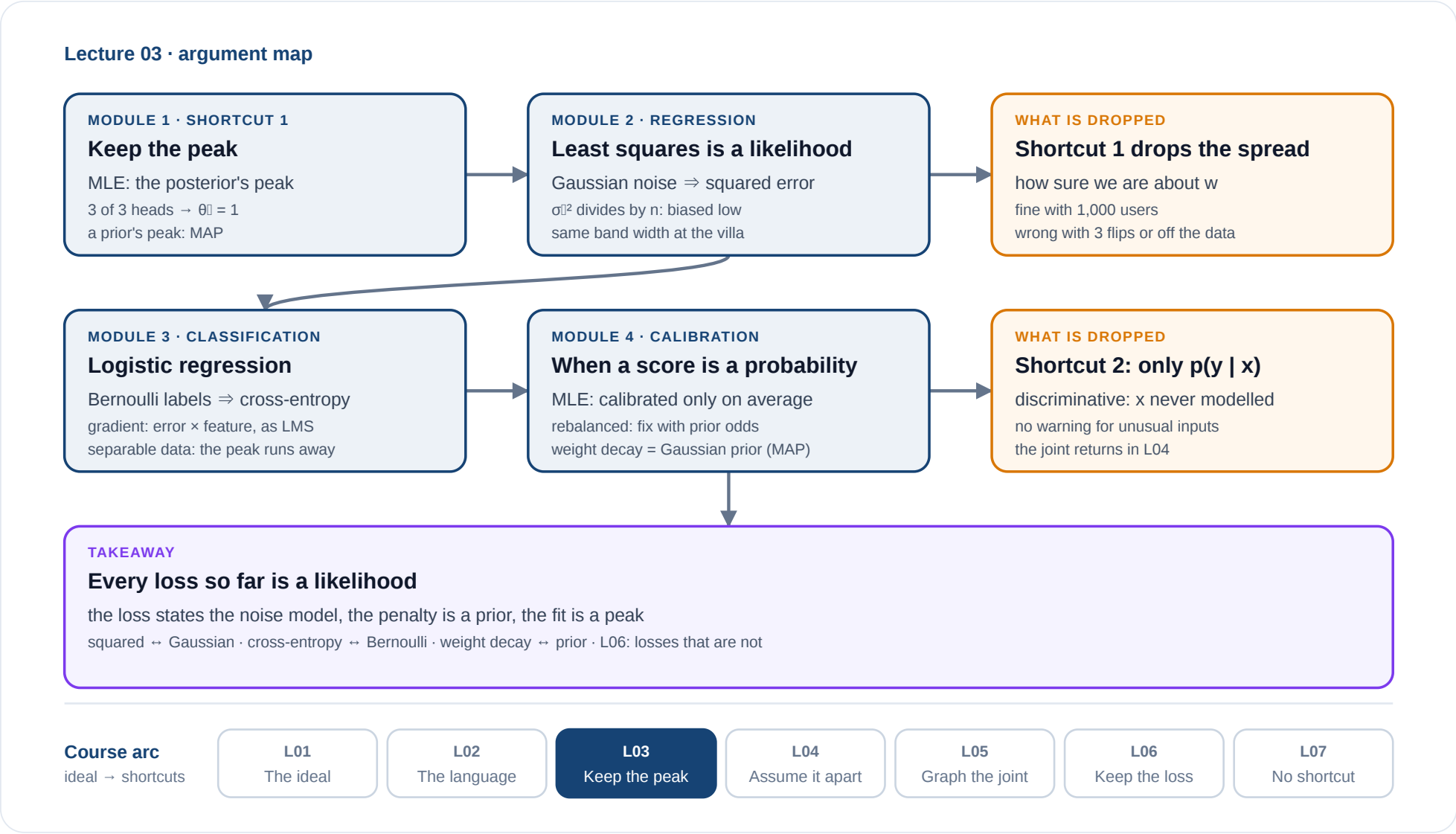
Lecture 01's table, rows 1 and 2 — methods you have used, read for what they drop:

shortcut	instead of the ideal	what is lost	where
Keep the peak	the most probable parameter — MLE or MAP	how sure we are about it	a coin · regression
Only $p(y \mid x)$	model the label given the features, never the joint — <i>discriminative</i>	every question about x	classification



The roadmap of the lecture in two forms: the two rows of the Lecture 01 table that today covers, and the four modules. Module 1 takes shortcut 1 where it can be drawn (the Beta posterior); Module 2 shows that least squares takes it, and what it costs far from the data; Module 3 takes shortcut 2 for classification; Module 4 answers Lecture 01's open question — when a score deserves the name probability.

Argument Map



Speaker notes

Top row: shortcut 1 — the peak on a coin, least squares as a likelihood, and what keeping the peak drops. Middle row: logistic regression as the Bernoulli twin, calibration and priors, and what modelling only $p(y | x)$ drops. Bottom: every loss so far is a likelihood. The strip marks Lecture 03's place in the arc. Return to the takeaway box at the end of the lecture.



Keep the Peak

Guiding Question: Last week's posterior gave a peak, a mean and an interval. What do we lose when we keep only the peak — and why does almost everyone do it?

The Likelihood, Read Along the Other Axis

The **likelihood** is the probability of the observed data, read as a function of the **parameter** with the data held fixed: $L(\theta) = p(D \mid \theta)$. The **maximum likelihood estimate** is its peak, $\hat{\theta}_{\text{ML}} = \arg \max_{\theta} p(D \mid \theta)$.

Lecture 02, 3 of 10 users

$$p(\theta \mid D) \propto \theta^3(1 - \theta)^7 \cdot 1$$

With a **flat prior** the posterior *is* the normalised likelihood — so its peak $h/n = 0.3$ is the maximum likelihood estimate.

Not a distribution over θ

$$\int_0^1 \theta^3(1 - \theta)^7 d\theta = \frac{1}{1320}$$

The likelihood is a probability over **data**. Read along θ it need not integrate to 1 — only the posterior does.

Speaker notes

Start from the curve the students drew last week, not from a new definition. The likelihood is the one ingredient of the posterior that the data supplies; maximum likelihood keeps that ingredient, drops the prior (or takes it flat) and then keeps only the highest point. The integral makes the warning concrete: $1/1320$ is $B(4, 8)$, the Beta normaliser — the number the posterior divides by. The warning is the most common confusion of the course: "the likelihood of θ " is not "the probability of θ ".



Deriving h/n Without a Posterior

Lecture 02's i.i.d. assumption lets the likelihood factorise over rows; the logarithm turns the product into a sum without moving the peak:

$$\ell(\theta) = \log \prod_{i=1}^n \theta^{y_i} (1 - \theta)^{1-y_i} = h \log \theta + (n - h) \log(1 - \theta)$$
$$\frac{d\ell}{d\theta} = \frac{h}{\theta} - \frac{n - h}{1 - \theta} = 0 \quad \implies \quad \hat{\theta}_{\text{ML}} = \frac{h}{n}$$

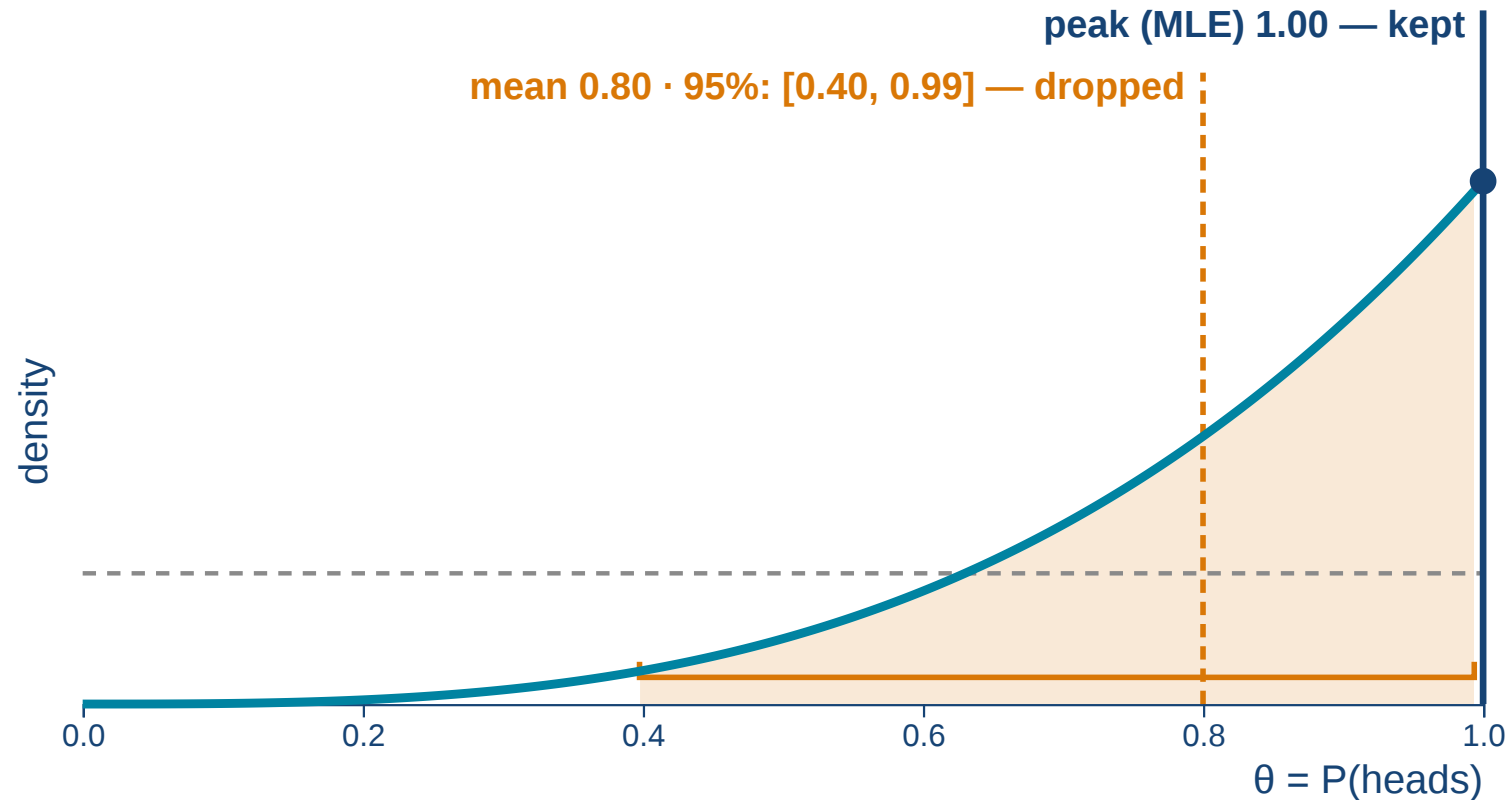
Last week's peak, found by **setting a derivative to zero** — no prior, no normaliser, no integral. The recipe works unchanged for a million parameters, where nobody can draw the posterior.

Math first: show the two lines and ask what they compute before saying it. The three reasons for the logarithm (same arg max, no underflow, sums differentiate term by term) can be said in one breath; students have met them in earlier ML courses. The insight box answers the second half of the guiding question: the peak is taken because it turns inference (an integral) into optimisation (a derivative). That is also why every method they know is fitted by gradient descent.

The Peak Can Lie

Three flips, **three heads**. Maximum likelihood: $\hat{\theta}_{\text{ML}} = 3/3 = 1$.

Flat prior (dashed), 3 heads: posterior Beta(4, 1)



Predict before the second click: where do the posterior mean and the 95% interval lie?

Speaker notes

Click 1 marks the peak — exactly 1: the estimate rules tails out forever. Stop and ask where the mean and the interval are. Click 2: mean 0.8, 95% interval [0.40, 0.99]. The posterior, with the same flat prior as last week, says "probably above a half, possibly well below 1"; the peak says "never tails". Nothing was wrong with the data or the model — only with keeping one number.

Numbers: Beta(4, 1) has density $4\theta^3$, CDF θ^4 , so the 2.5% and 97.5% points are $0.025^{(1/4)} = 0.398$ and $0.975^{(1/4)} = 0.994$.



A Prior's Peak: MAP

Keep the prior, and still take only the peak — the **maximum a posteriori** estimate. With a $\text{Beta}(a, b)$ prior, the same derivative gives

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(D \mid \theta)p(\theta) = \frac{h + a - 1}{n + a + b - 2}$$

prior	data	MLE h/n	MAP	posterior mean
Beta(1, 1) flat	3 of 3	1	1	0.80
Beta(2, 2)	3 of 3	1	0.80	0.71
Beta(2, 2)	8 of 10 (Lecture 02)	0.80	0.75	0.71

The prior shifts the peak as if $a - 1$ yeses and $b - 1$ noes had been seen before the data — a **flat** prior shifts nothing. MAP pulls 3 of 3 back to 0.8 — and still reports one number.

Speaker notes

The last row is the Lecture 02 mini-exercise (posterior $\text{Beta}(10, 4)$): students computed a MAP then without the name. Derivation for the curious: add $(a - 1) \log \theta + (b - 1) \log(1 - \theta)$ to $\ell(\theta)$ and set the derivative to zero. MAP is the bridge to Module 4, where a penalty on the weights turns out to be exactly this: a prior's peak. Note that the posterior mean $(a + h)/(a + b + n)$ is a different point — the shortcut chooses the mode, and for a skewed posterior the two differ.



Why Everyone Keeps the Peak

	the whole posterior	only the peak
needs	the normaliser $p(D)$ — an integral over θ	a derivative
in 1 dimension	a Beta table	h/n
in a million	an integral nobody can compute exactly	gradient descent
costs	computation	the spread

With **1,000 users** the posterior is Beta(301, 701): standard deviation 0.015. The peak is nearly everything there is to know.

Speaker notes

The trade in one table: the posterior asks for an integral, the peak for a derivative. With plenty of data the posterior is so narrow (Lecture 02: [0.27, 0.33]) that its peak is an excellent summary; with three flips it is not. The shortcut is not wrong, it is a bet that the spread does not matter — Module 2 shows the place where it matters even with plenty of data: far from the data.



Mini-Exercise

Two analysts both report $\hat{\theta} = 0.3$. One saw **3 of 10** users return, the other **300 of 1,000**.

What does the reported number hide — and which estimate would you bet on?

One minute to read, one to think. Everything needed is on last week's slide; the arithmetic is not the point. Standard deviations 0.13 and 0.015 (ratio about 9).

Mini-Exercise

Two analysts both report $\hat{\theta} = 0.3$. One saw **3 of 10** users return, the other **300 of 1,000**.

What does the reported number hide — and which estimate would you bet on?

The **spread**. Lecture 02's posteriors: $[0.11, 0.61]$ after 10 users, $[0.27, 0.33]$ after 1,000 — the same peak, an interval nine times wider. Bet on the second; a point estimate cannot tell you to.

Takeaway

Shortcut 1 — keep the peak. Maximum likelihood (flat prior) and MAP (with a prior) keep the posterior's most probable point. It is cheap — a derivative instead of an integral — and harmless with plenty of data. It drops the **spread**: how sure we are about the parameter.

Least Squares Is a Likelihood

Guiding Question: You have minimised squared error many times. What did that choice assume about the world — and what does it leave out?

Linear Regression in One Slide

A price model for flats of **40–120 m²**, with $x_0 = 1$ for the intercept — everything else is in the backup:

piece	recap
model	$\hat{y} = \mathbf{w}^\top \mathbf{x} = w_0 + w_1 x_1 + \cdots + w_d x_d$
objective	$\text{RSS}(\mathbf{w}) = \sum_{i=1}^n (y_i - \mathbf{w}^\top \mathbf{x}_i)^2$
closed form	normal equations: $\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$
iterative	LMS, one example at a time: $\mathbf{w} \leftarrow \mathbf{w} + \eta(y_i - \hat{y}_i)\mathbf{x}_i$ — error $\times$ feature
justified by	Gauss–Markov: best <i>linear unbiased</i> estimator if the errors have mean zero, equal variance and no correlation

Speaker notes

A recap, not a lesson: students have fitted this model in earlier courses. Say the one sentence that matters for later: LMS is gradient descent on RSS, example by example, and its update is error times feature — the same update returns for logistic regression in Module 3. The Gauss–Markov line is the classical justification; it says "best among linear unbiased estimators" and needs no distribution. Keep it in mind as the contrast for the next slides. The closed form for one feature, R^2 , the geometry and learning rates are in the backup at the end of the deck.



The Same Model, Told Differently

Classical account

Choose the line that **minimises squared error**:

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w}} \sum_i (y_i - \mathbf{w}^\top \mathbf{x}_i)^2$$

Probabilistic account

State how a price **arises** for a given flat:

$$y = \mathbf{w}^\top \mathbf{x} + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2)$$

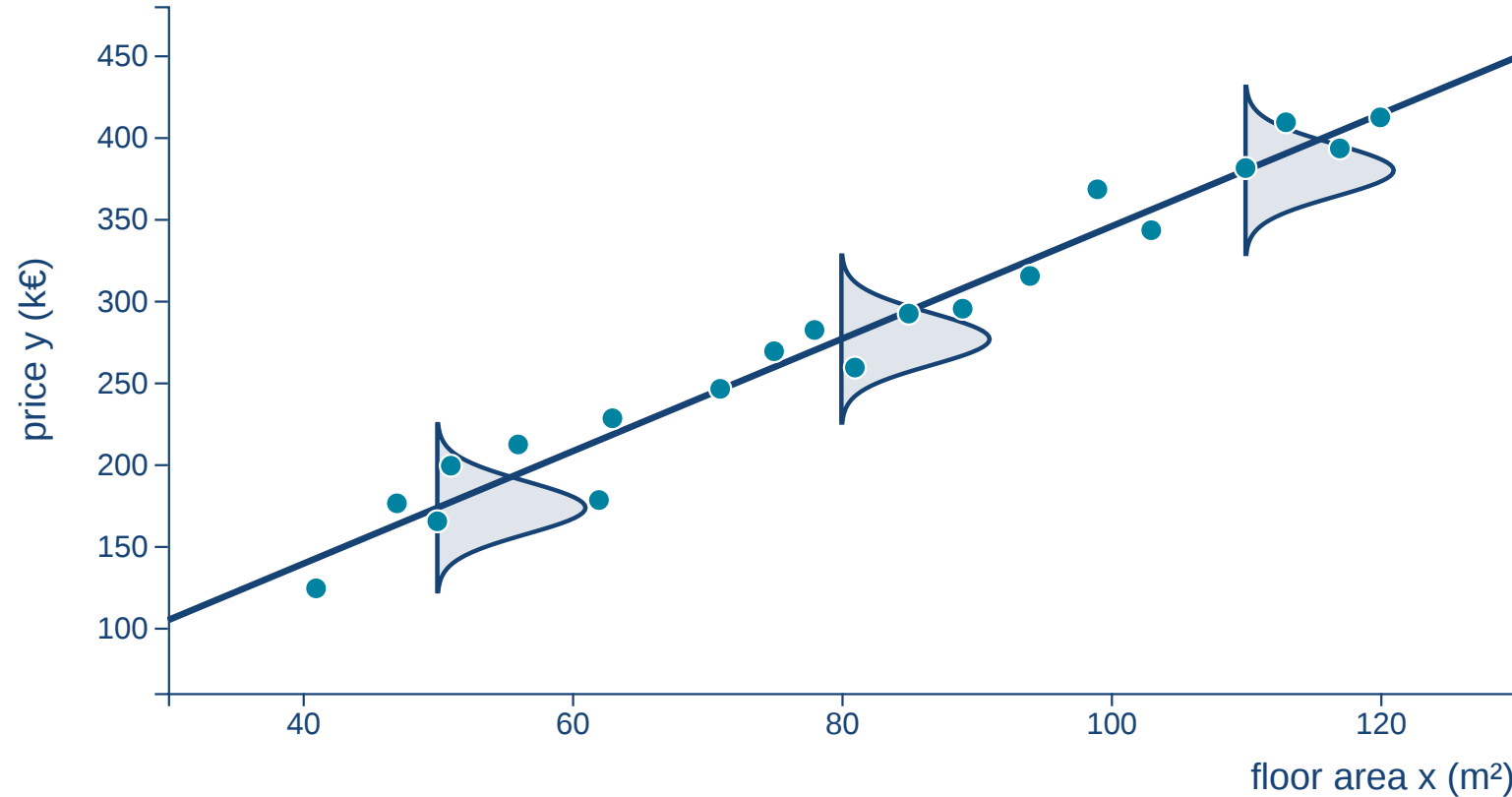
then keep the peak of the likelihood.

Do the two accounts give the same $\hat{\mathbf{w}}$ — and if so, what did the classical account assume without saying it?

Question first (P1). Let students vote. Most will guess "same" without being able to say why; the next two slides give the picture and then the derivation. The second half of the question is the module's guiding question in small.

A Distribution at Every x

$p(y | x) = \mathcal{N}(\hat{w}_0 + \hat{w}_1 x, \sigma^2)$, $\sigma = 16 \text{ k€}$ — the same bell at every x



A **conditional distribution** $p(y | x) = \mathcal{N}(\mathbf{w}^\top \mathbf{x}, \sigma^2)$: one bell per x , **equally wide**. Nothing about how x arises.

Speaker notes

Click 1: the fitted line, $\hat{y} \approx 1 + 3.44 \cdot x$ (k€, m²). Click 2: the bells at 50, 80 and 110 m², each with $\sigma \approx 16$ k€. Read a bell aloud: "an 80 m² flat costs about 276 k€, give or take 30". This is the picture that makes both later points of the lecture visible: the loss is the shape of the bell (next slide), and the model is only about *y given x* — there is no bell for *x* itself. That second point is shortcut 2, named in Module 3: the model never learns what a typical flat looks like, so it cannot tell that a villa is unusual.



The Derivation

With $p(y_i \mid \mathbf{x}_i, \mathbf{w}) = \mathcal{N}(y_i \mid \mathbf{w}^\top \mathbf{x}_i, \sigma^2)$ and i.i.d. rows:

$$\ell(\mathbf{w}) = \sum_{i=1}^n \log \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(y_i - \mathbf{w}^\top \mathbf{x}_i)^2}{2\sigma^2} \right) = \underbrace{-\frac{n}{2} \log(2\pi\sigma^2)}_{\text{no } \mathbf{w}} - \frac{1}{2\sigma^2} \text{RSS}(\mathbf{w})$$

Maximising the Gaussian log-likelihood **is** minimising RSS: $\hat{\mathbf{w}}_{\text{ML}} = \hat{\mathbf{w}}_{\text{OLS}}$. Squared error is the exponent of the bell.

Speaker notes

Walk the line slowly — the course's thesis becomes concrete here. The first term does not contain w , the factor $1/(2\sigma^2)$ is positive, so the $\arg \max$ over w is the $\arg \min$ of RSS whatever σ is. Answer to the previous slide's vote: the same $\hat{w}$ — and the classical account had assumed, without saying so, Gaussian noise of one fixed width.



What Squared Error Assumed

the noise model says	squared error silently assumes	check on the residuals
ε has mean 0	the line is the right shape	no pattern against x
one σ for every x	homoscedastic noise	no fan shape
rows independent	i.i.d. (Lecture 02)	no pattern in time or space
ε Gaussian	thin tails: large errors are rare	no heavy tails, few outliers

Choosing a loss and choosing a noise model are **one decision**. Squared error $\Leftrightarrow$ Gaussian noise; absolute error $\Leftrightarrow$ Laplace noise — heavier tails, so a large residual is less surprising and an outlier pulls the fit less.

Each row turns a property of the loss into a claim about the world, and every claim can be checked with a residual plot — that is what stating the model buys. The Laplace row answers a question students may bring from earlier courses ("why is L1 loss robust?") with a reason instead of a rule: $-\log$ of a Laplace density is $|\epsilon|/b$ plus a constant.



The Peak of σ^2

The same likelihood also has a peak in σ^2 :

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\text{RSS}}{2\sigma^4} = 0 \quad \implies \quad \hat{\sigma}_{\text{ML}}^2 = \frac{\text{RSS}}{n}$$

Which of Lecture 02's two variances is this — and is it too small or too large?

Speaker notes

A second, quiet cost of keeping the peak. Lecture 02 divided by $n - 1$ because one mean had been estimated; here p weights have been, so $E[\text{RSS}] = (n - p)\sigma^2$ and $E[\sigma^2_{\text{ML}}] = (n - p)/n \cdot \sigma^2$. With 20 flats that is 10% in the variance; with 5 flats and 2 weights it would be 40%. Small or not, it always points the same way: the peak makes the model look less noisy than it is.



The Peak of σ^2

The same likelihood also has a peak in σ^2 :

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\text{RSS}}{2\sigma^4} = 0 \quad \implies \quad \hat{\sigma}_{\text{ML}}^2 = \frac{\text{RSS}}{n}$$

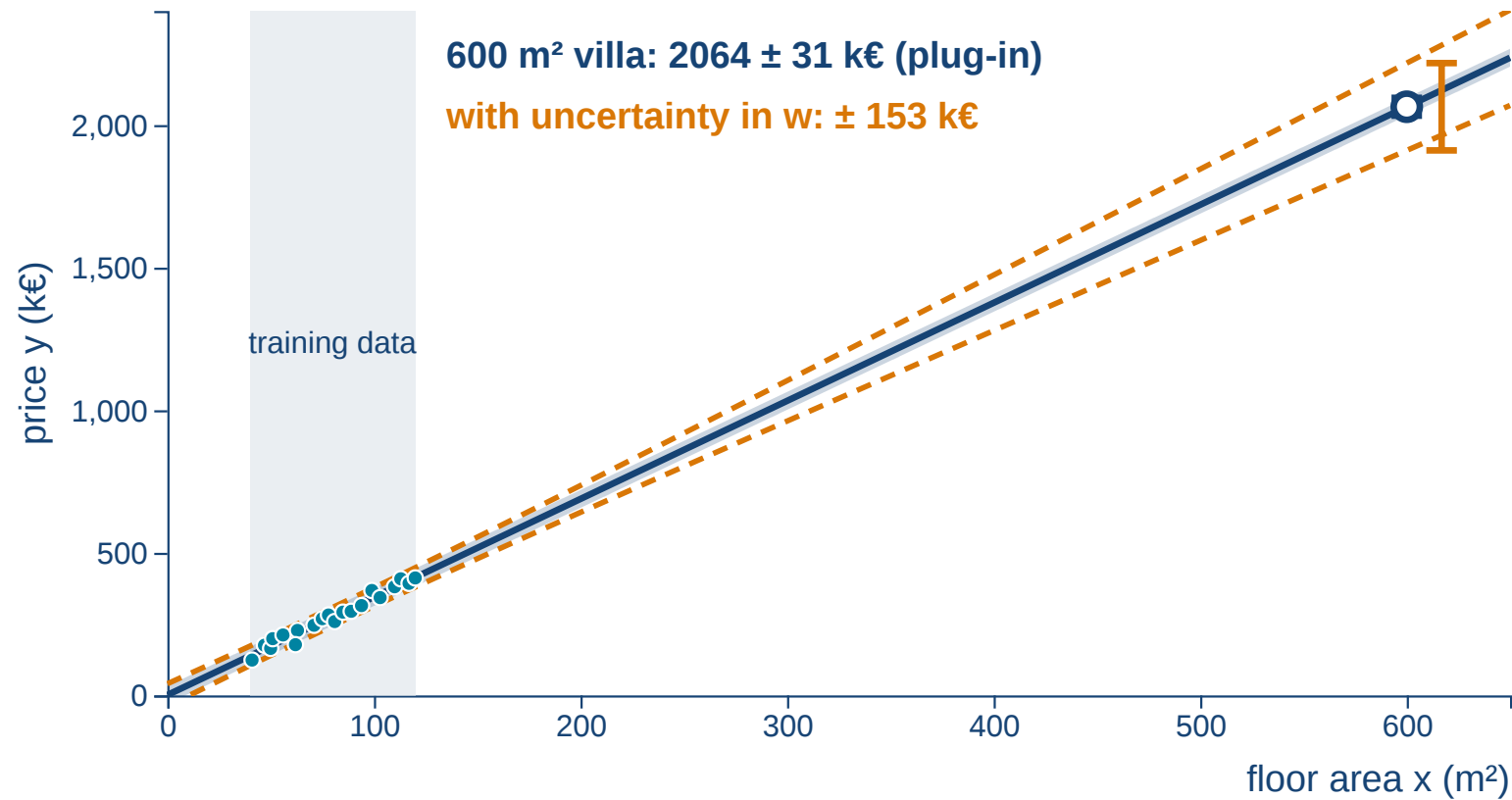
Which of Lecture 02's two variances is this — and is it too small or too large?

The $1/n$ one. The line was fitted to the same data, so residuals are smaller than the true errors: $\hat{\sigma}_{\text{ML}}^2$ is **biased low**. The unbiased version divides by $n - p$ for p fitted weights.

Houses: $n = 20$, $p = 2$, $\text{RSS} \approx 5,097 \rightarrow \hat{\sigma} \approx 16.0$ instead of 16.8 k€.

The Villa Again

The peak dropped the uncertainty about w — it grows off the data



The band is as narrow at the villa as at a flat. Is the model as sure?

Speaker notes

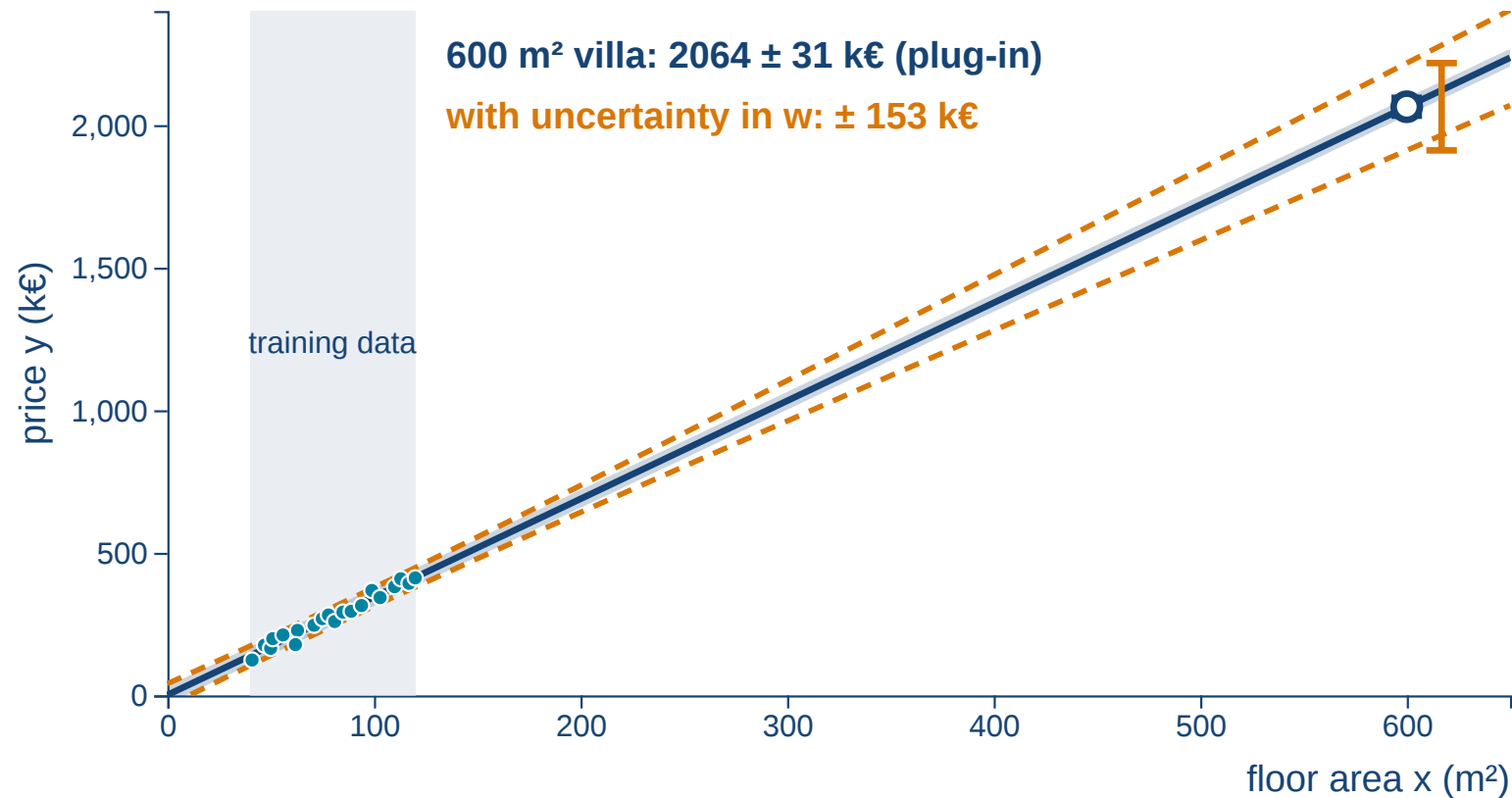
Step 0: the fit and the plug-in band $\hat{y} \pm 1.96 \cdot \sigma = \pm 31$ k€. Click 1: the villa of Lecture 01 — predicted about 2,064 k€ ± 31 k€, the same half-width as for any flat. Ask the question before the second click. Click 2: the band that also counts the uncertainty about the two weights, $1.96 \cdot \sigma \cdot \sqrt{(1 + 1/n + (x - \bar{x})^2 / \sum (x_i - \bar{x})^2)}$: ± 32 k€ at 80 m², ± 153 k€ at 600 m². This is Lecture 01's epistemic uncertainty, made visible — and exactly the spread that Module 1 said the peak drops.

One more honesty point, for the discussion: even the wide band trusts the straight line all the way to 600 m². Whether villas follow the price per m² of flats is a modelling question no band can answer.



The Villa Again

The peak dropped the uncertainty about w — it grows off the data



The band is as narrow at the villa as at a flat. Is the model as sure?

No: ± 31 k€ is only the noise $\hat{\sigma}$. The uncertainty about w — dropped with the peak — grows off the data: ± 153 k€ at the villa (Lecture 07).

Mini-Exercise

The residuals of your house-price model fan out: about ± 10 k€ for 40 m² flats, ± 60 k€ for 120 m² flats.

Which assumption fails — and which output of the model can you no longer trust?

Speaker notes

One minute to read, one to think. Students with a statistics background may add that $\hat{w}$ stays unbiased but is no longer the most efficient estimator (the Gauss–Markov condition "equal variance" fails) — correct, and worth one sentence. The fix, if asked: model σ as a function of x , which is again just stating a different noise model.



Mini-Exercise

The residuals of your house-price model fan out: about ± 10 k€ for 40 m² flats, ± 60 k€ for 120 m² flats.

Which assumption fails — and which output of the model can you no longer trust?

One σ for every x (homoscedasticity). The line may still be fine, but the **band** is not: a single $\hat{\sigma}$ is too wide for small flats and too narrow for large ones.

Takeaway

Least squares **is** maximum likelihood under Gaussian noise: the loss states the noise model, and the model is a distribution $p(y \mid x)$. It is also a **peak**: $\hat{\sigma}^2$ divides by n , and the band is as narrow at the villa as at a flat — the uncertainty about $\mathbf{w}$ was dropped.

Classification: Model Only $p(y | x)$

Guiding Question: Lecture 01's fraud model said 0.55. Where does such a number come from — and what does the model refuse to know?

Why Not Regress on 0/1?

Code fraud as $y = 1$, legitimate as $y = 0$, and fit Module 2's line.

Speaker notes

A known failure, stated in the language of Module 2: the loss is wrong because the noise model is wrong. A Gaussian around $w^T x$ is a model of a real number; fraud is a yes/no outcome. Everything else in the module follows from replacing the Gaussian by the Bernoulli.



Why Not Regress on 0/1?

Code fraud as $y = 1$, legitimate as $y = 0$, and fit Module 2's line.

Squared error punishes being right

A transaction far on the correct side has a large residual, so least squares drags the boundary towards cases it already classifies correctly.

The output is not a probability

$\mathbf{w}^\top \mathbf{x}$ can be 17 or -40 . The Gaussian bell of Module 2 puts mass on every real y — for a label that is 0 or 1.

Keep the linear score, change what it models: the **log-odds**, and a noise model that produces 0s and 1s.

Log-Odds Are Linear

$$\ln \frac{P(y = 1 \mid \mathbf{x})}{1 - P(y = 1 \mid \mathbf{x})} = \mathbf{w}^\top \mathbf{x} \quad \Longleftrightarrow \quad P(y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^\top \mathbf{x}}}$$

read the weights as

the intercept: the log-odds when every feature is 0

a weight: what one unit of a feature **adds** to the log-odds

the boundary: where $\mathbf{w}^\top \mathbf{x} = 0$

fraud example

no features at all: $w_0 = \ln \frac{1}{999} \approx -6.9$, the base rate 1 in 1,000

"new device" with $w = \ln 6 \approx 1.8$: odds $\times 6$

score 0.5 — still a straight line

Speaker notes

The sigmoid is not an arbitrary squashing function; it is the inverse of the logit. The reading of the intercept ties the model to Lecture 02: a fraud model with no features has its peak at the base rate (h/n again: 1 in 1,000, log-odds $\ln(1/999)$). The weight reading is the one to stabilise: weights add log-odds, so they multiply odds — the same bookkeeping as likelihood ratios in Bayes' theorem, which Lecture 04 will exploit.



Where Does Cross-Entropy Come From?

$$L(\mathbf{w}) = - \sum_{i=1}^n \left[y_i \ln h_i + (1 - y_i) \ln(1 - h_i) \right], \quad h_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$$

You have minimised this loss many times. Which noise model does it state?

Speaker notes

Math first (P5): the loss on the screen, the question, then the reveal. It is the same derivation as h/n in Module 1, with θ replaced by $h(x)$ for every row. Stress the last sentence: this is a *conditional* story, y given x . It is not a generative story — nothing in it could produce a transaction.



Where Does Cross-Entropy Come From?

$$L(\mathbf{w}) = - \sum_{i=1}^n \left[y_i \ln h_i + (1 - y_i) \ln(1 - h_i) \right], \quad h_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$$

You have minimised this loss many times. Which noise model does it state?

Given $\mathbf{x}$, the label is a **Bernoulli** draw with bias h : $P(y_i \mid \mathbf{x}_i, \mathbf{w}) = h_i^{y_i} (1 - h_i)^{1-y_i}$. With i.i.d. rows,

$$\ell(\mathbf{w}) = \sum_{i=1}^n \left[y_i \ln h_i + (1 - y_i) \ln(1 - h_i) \right] = -L(\mathbf{w})$$

Cross-entropy **is** the Bernoulli negative log-likelihood — Module 1's coin, biased by $\mathbf{x}$. How $\mathbf{x}$ arises is never said.

The Same Update Again

With $\sigma' = \sigma(1 - \sigma)$, the gradient of the log-likelihood is

$$\frac{\partial \ell}{\partial \mathbf{w}} = \sum_{i=1}^n (y_i - h_i) \mathbf{x}_i$$

model	noise	loss	gradient of ℓ
linear regression	Gaussian	squared error	$\frac{1}{\sigma^2} \sum_i (y_i - \mathbf{w}^\top \mathbf{x}_i) \mathbf{x}_i$
logistic regression	Bernoulli	cross-entropy	$\sum_i (y_i - \sigma(\mathbf{w}^\top \mathbf{x}_i)) \mathbf{x}_i$

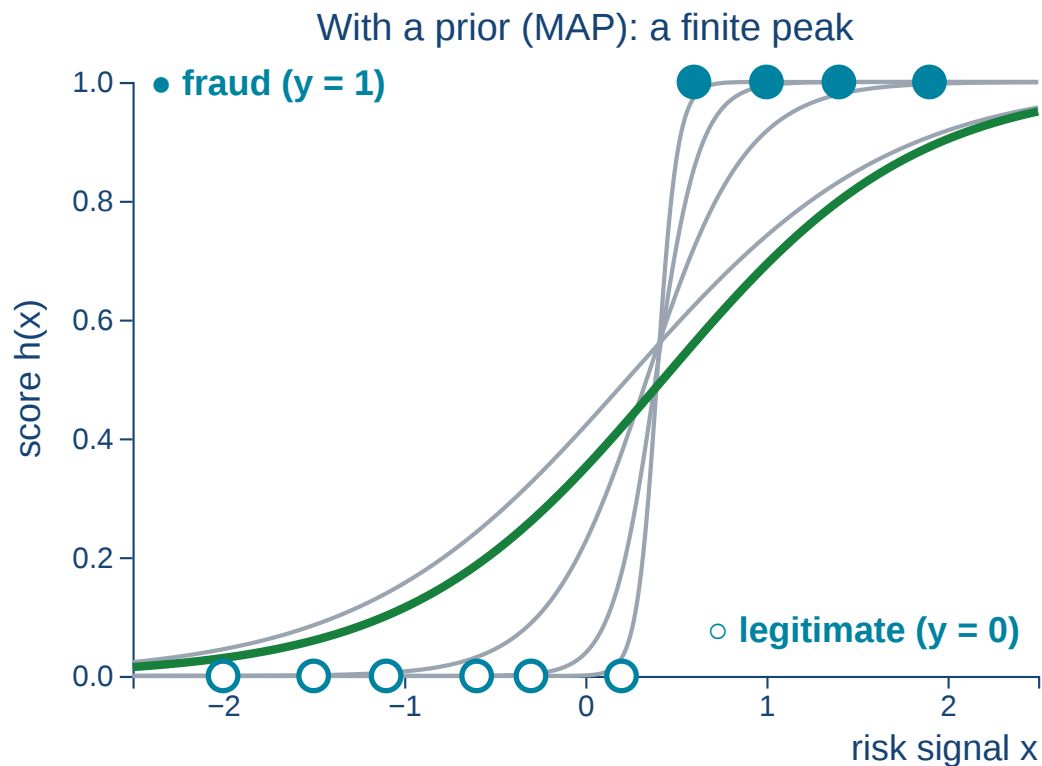
Error $\times$ feature — the LMS update with a sigmoid in the prediction; no closed form, so iterate. A fraud scored 0.01 costs $-\ln 0.01 \approx 4.6$, scored 0.99 only 0.01: confident mistakes dominate.

Speaker notes

One line of calculus: $d/dw [y \ln h + (1 - y) \ln(1 - h)] = (y/h - (1 - y)/(1 - h)) \cdot h(1 - h) \cdot x = (y - h) x$. The table is the unification the lecture is after: two noise models, one form of update. The per-example version is exactly LMS with $\hat{y}$ replaced by $\sigma(w^T x)$.



When the Peak Runs Away



steps	slope	log-lik
10	1.4	-2.68
100	3.6	-1.04
1,000	8.3	-0.36
10,000	17.8	-0.06
MAP	1.4	-2.54
prior on the slope: $N(0, 1)$ stays, however long it runs		

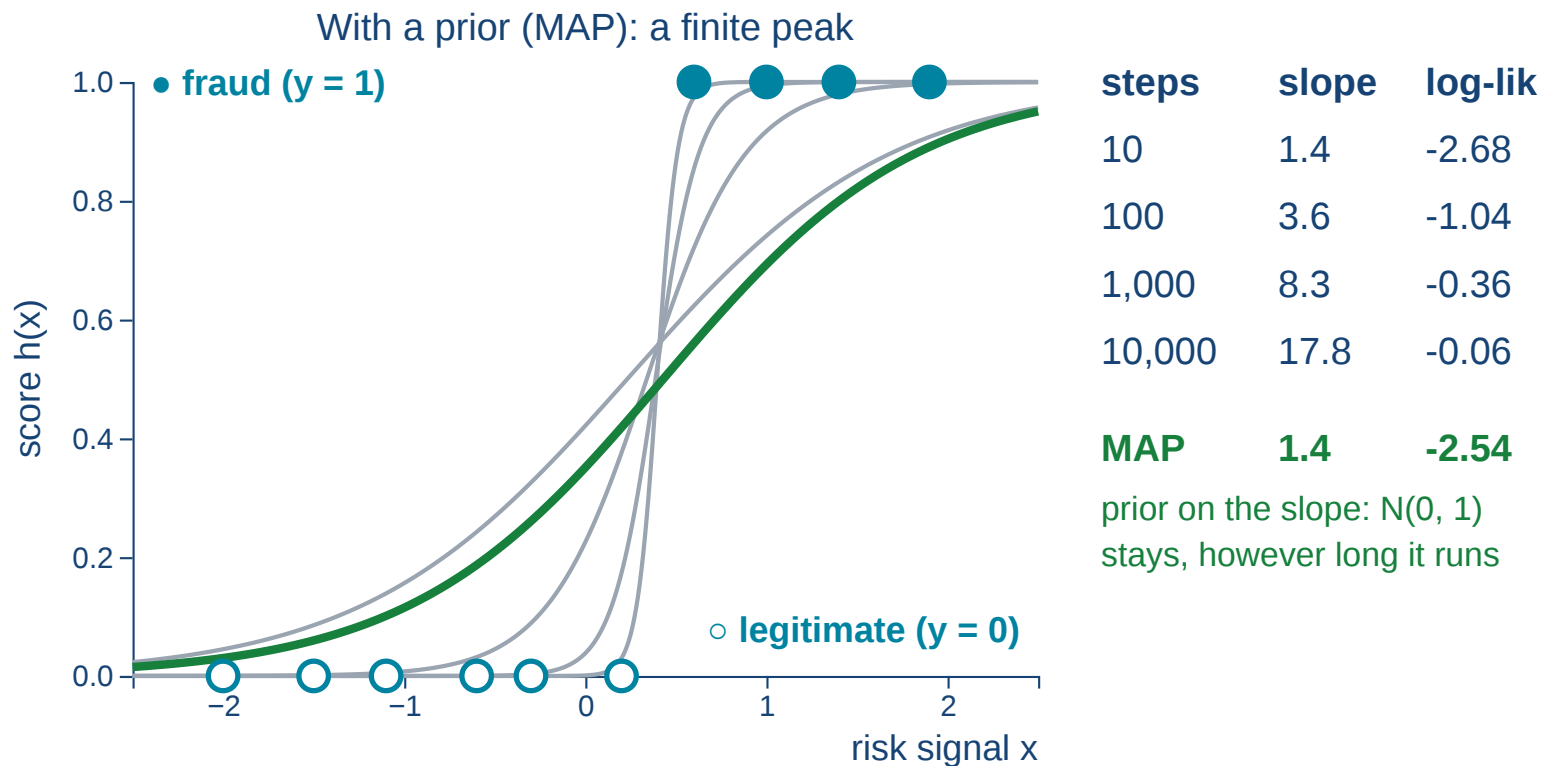
Every fraud scores above every legitimate case. Where is the peak of the likelihood?

Speaker notes

Ask the question before the first click and let students guess. Clicks 1–4: the curve after 10, 100, 1,000 and 10,000 gradient steps — slope 1.4, 3.6, 8.3, 17.8, log-likelihood -2.68 , -1.04 , -0.36 , -0.06 . It never stops; the optimiser is not broken, the peak does not exist. Click 5: a Gaussian prior $N(0, 1)$ on the slope; MAP converges to slope 1.4 and stays there. In practice every library adds this prior by default — it is called regularisation, and Module 4 reads it as a prior. Separable data is common with many features (rare fraud, thousands of attributes), so this is not an exotic case.



When the Peak Runs Away



Every fraud scores above every legitimate case. Where is the peak of the likelihood?

Nowhere. The likelihood keeps rising as the slope grows; the scores harden to 0 and 1 — three heads in three flips. A prior's peak (MAP) is finite.

Shortcut 2: Model Only $p(y \mid \mathbf{x})$

Linear and logistic regression are **discriminative**: a model of y given $\mathbf{x}$, none of $\mathbf{x}$.

What is dropped: $p(\mathbf{x})$

- a transaction unlike any seen still gets a confident score
- a missing field cannot be averaged out — it must be filled in
- no way to simulate a plausible transaction

What is gained

- no assumptions about how features are distributed — where, with hundreds of features, most modelling errors hide
- every parameter serves the decision

Lecture 04 takes the other route: model the joint $p(\mathbf{x}, y)$ and condition — **generative**.

Name the shortcut explicitly and tie it to the L01 roadmap row "only what you need". The first dropped item is the villa of Module 2 in classification form: the model cannot know that an input is unusual, because it never modelled inputs. The gained column is not a consolation prize: for prediction with many features, discriminative models are usually more accurate — Lecture 04 makes the comparison with naive Bayes.

Mini-Exercise

Your fraud model scores **0.97** for a transfer 50 times larger than any in its training data.

Should you trust the 0.97 — and could the model itself warn you?

Speaker notes

One minute to read, one to think. The transfer is the classification twin of the 600 m² villa. Good follow-up if time allows: "which of Lecture 01's two kinds of uncertainty is this?" — epistemic; ten such transfers in the training data would help.



Mini-Exercise

Your fraud model scores **0.97** for a transfer 50 times larger than any in its training data.

Should you trust the 0.97 — and could the model itself warn you?

Not blindly: the linear log-odds simply **extrapolate**, like the line at the villa. And no — the model has **no** $p(\mathbf{x})$, so nothing in it can see that the amount is unusual. That needs a model of $\mathbf{x}$ or a separate check.

Takeaway

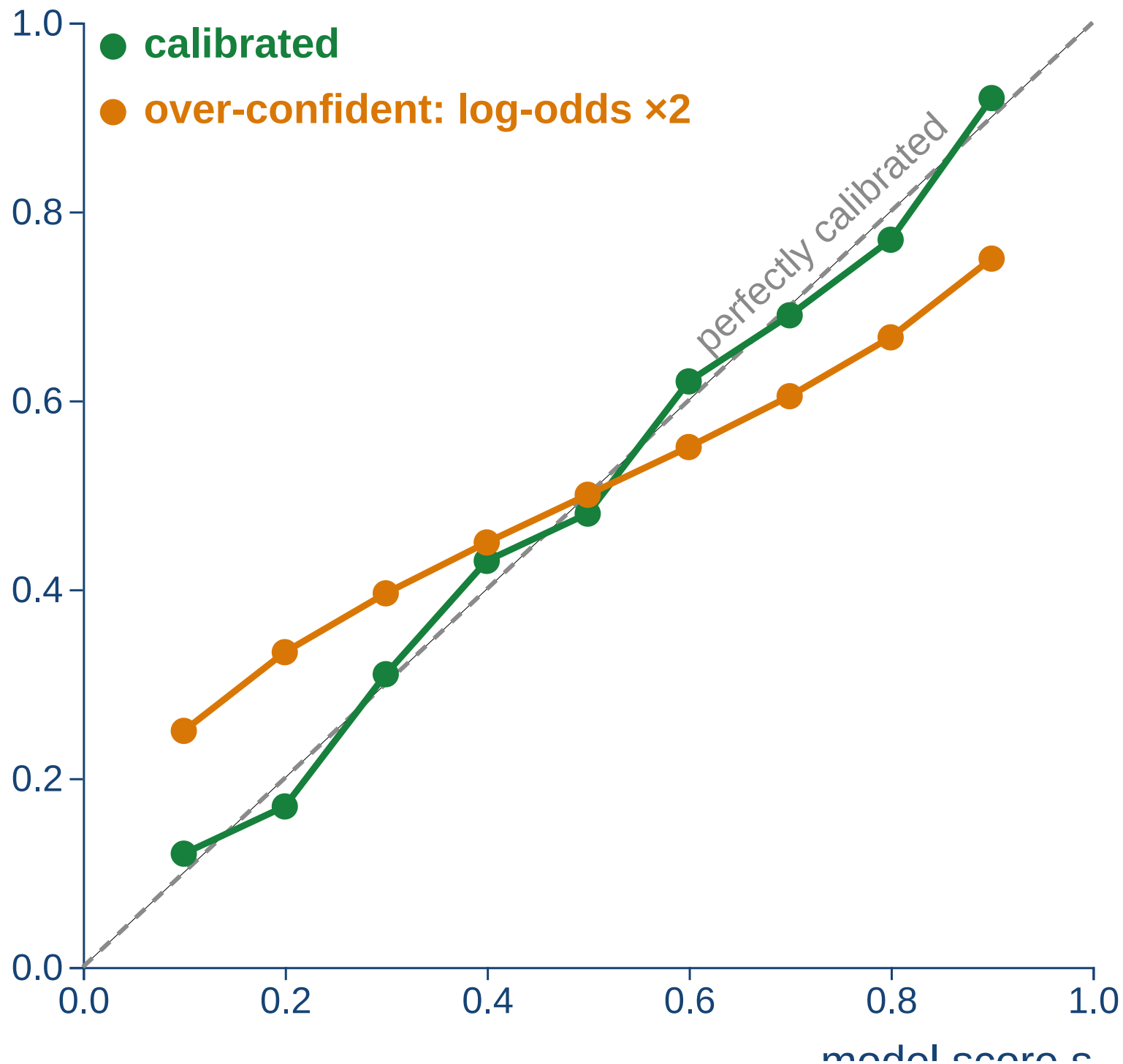
Logistic regression is Module 2 with Bernoulli noise: linear log-odds, cross-entropy as the negative log-likelihood, error $\times$ feature as the gradient. It **keeps the peak** — which runs away on separable data — and **models only** $p(y \mid \mathbf{x})$: it never learns what a normal transaction looks like.

When Is a Score a Probability?

Guiding Question: Lecture 01 promised to say when a model's output earns the name "probability". When does it — and what breaks it?

When Is 0.7 a Probability?





A score is **calibrated** if, among all cases scored s , a fraction s is positive. A **reliability diagram** plots that fraction against s .

Predict before click 2: a model whose log-odds are twice too large — above or

below the diagonal at high
scores?

Step 0: the diagonal. Click 1: a calibrated model — the points scatter around the diagonal, as bins of real data would. Ask for the prediction, then click 2: the over-confident model lies *below* the diagonal at high scores and *above* it at low ones — its 0.9 is really 0.75, its 0.1 really 0.25. Lecture 04 shows a model that is over-confident by construction (naive Bayes counting correlated evidence twice). Only a calibrated score can go into Lecture 01's cost threshold $p^* = 0.1$; a 0.9 that means 0.75 moves every decision near the cut.

What Maximum Likelihood Guarantees

At the peak, the gradient for the intercept ($x_0 = 1$) is zero:

$$\sum_{i=1}^n (y_i - h_i) = 0 \quad \implies \quad \frac{1}{n} \sum_{i=1}^n h_i = \frac{1}{n} \sum_{i=1}^n y_i$$

What does this equation promise — and what does it not?

Speaker notes

Math first: the equation is Module 3's gradient, read for the intercept only. It is a genuine guarantee, but a weak one: it holds on average and on the data the model was fitted to. Per-score calibration needs the model to be right (correctly specified), which a linear log-odds rarely is exactly; in practice logistic regression is often close, and that is why it has a reputation for "calibrated probabilities" — a reputation, not a theorem. Separation (ranking fraud above legitimate cases) is a different property, measured in Lecture 06.



What Maximum Likelihood Guarantees

At the peak, the gradient for the intercept ($x_0 = 1$) is zero:

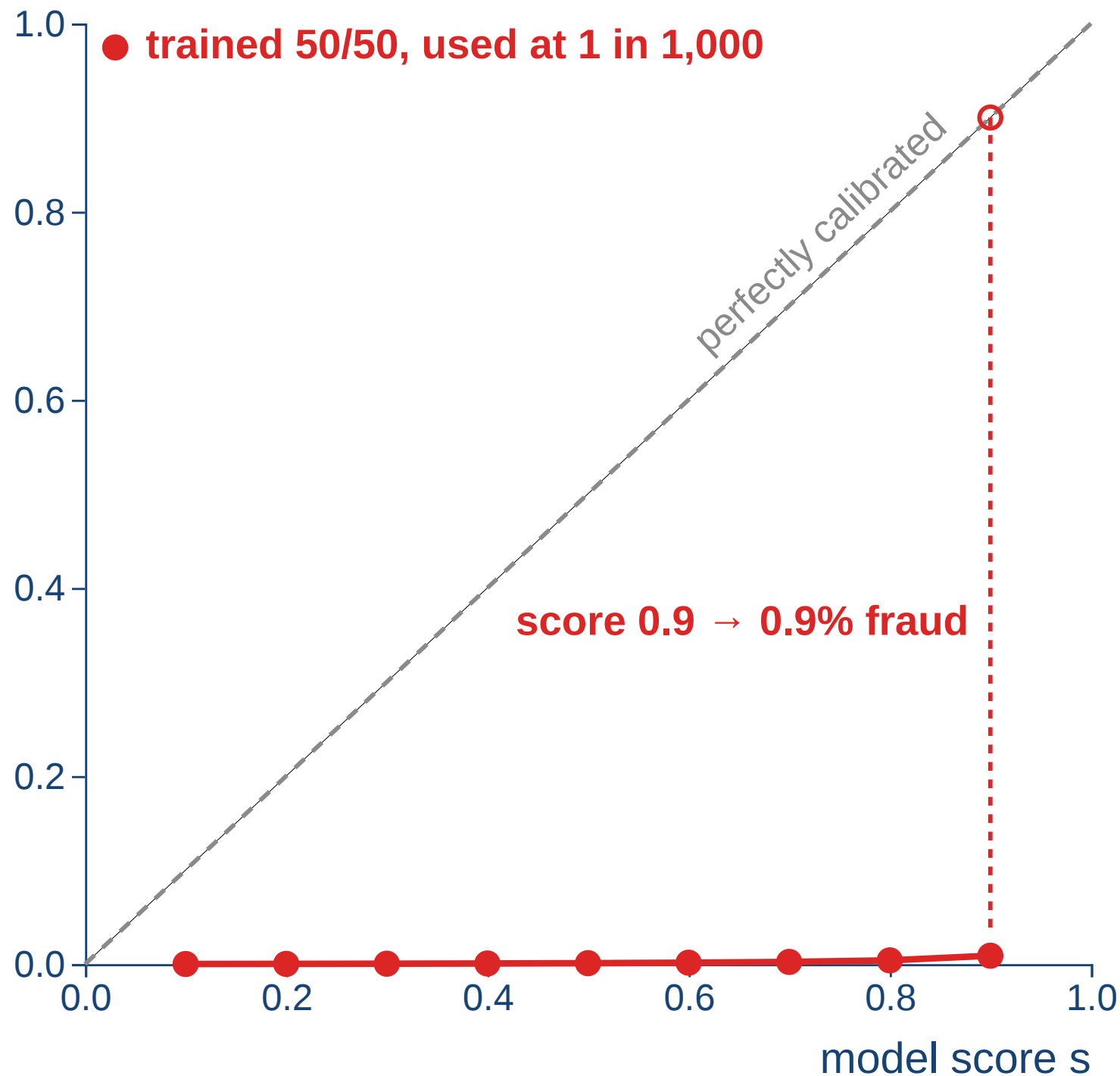
$$\sum_{i=1}^n (y_i - h_i) = 0 \implies \frac{1}{n} \sum_{i=1}^n h_i = \frac{1}{n} \sum_{i=1}^n y_i$$

What does this equation promise — and what does it not?

promised	not promised
average score = training fraud rate: <i>calibration in the large</i>	each score right — needs log-odds truly linear in $\mathbf{x}$
on the training distribution	after a base-rate shift, or rebalancing
	that it separates : the base rate for everyone is calibrated — and useless

Rebalanced Training





Fraud is 1 in 1,000, so practitioners **undersample** legitimate cases to 50/50. The model learns odds for a world where fraud is common.

Bayes in odds form:

$$\text{real odds} = \text{model odds} \times \frac{1/999}{1}$$

Score 0.9: odds 9 → 9/999, about **0.9%**.

Speaker notes

The click shows the rebalanced model on real traffic: every point hugs the floor. A score of 0.9 means about 0.9% fraud (odds 9/999, probability $9/1008 = 0.89\%$). The repair is Bayes' theorem from Lecture 02 in odds form: undersampling legitimate cases changes the prior, not $p(x | y)$, so the model's odds are the likelihood ratio times the *training* prior odds; swap in the real prior odds. General form: real odds = model odds $\times$ (real prior odds) / (training prior odds); here $(1/999) / 1$. The base rate that made a "99% accurate" flag mean 9% in Lecture 02 returns — and a threshold of 0.5 on the raw score would block almost only honest customers.



Regularisation Is a Prior

$$\min_{\mathbf{w}} \text{RSS}(\mathbf{w}) + \lambda \|\mathbf{w}\|^2$$

You know this as ridge regression or weight decay. Which prior's peak is it?

Multiply the log-posterior by $-2\sigma^2$ to get the penalised objective; $\lambda = \sigma^2/\tau^2$ reads well: noisy data (large σ^2) or a tight prior (small τ^2) means more shrinkage. This is Lecture 02's Beta(2, 2) exercise one level up: the prior pulled the peak from 0.8 to 0.75, the Gaussian prior pulls weights towards 0. For logistic regression the same prior gives cross-entropy + $\|w\|^2/(2\tau^2)$ — the finite peak of the runaway slide. The intercept is usually left unpenalised, so calibration in the large survives. L1: the Laplace log-density $-|w|/b$ has a corner at 0, and at a corner the peak can sit exactly at 0 over a whole range of data; averaging over the posterior smooths the corner away.

Regularisation Is a Prior

$$\min_{\mathbf{w}} \text{RSS}(\mathbf{w}) + \lambda \|\mathbf{w}\|^2$$

You know this as ridge regression or weight decay. Which prior's peak is it?

MAP from Module 1 with $\mathbf{w} \sim \mathcal{N}(0, \tau^2 \mathbf{I})$: $\log p(\mathbf{w} \mid D) = -\frac{\text{RSS}}{2\sigma^2} - \frac{\|\mathbf{w}\|^2}{2\tau^2} + \text{const}$, so $\lambda = \sigma^2 / \tau^2$.

penalty	prior on $\mathbf{w}$	what its peak does
$\lambda \ \mathbf{w}\ ^2$	Gaussian	shrinks; keeps separable data finite
$\lambda \ \mathbf{w}\ _1$	Laplace	exact zeros — a property of the peak (the kink at 0); the posterior mean has none

Mini-Exercise

A fraud model trained on a **50/50** sample is used on real traffic, where fraud is **1 in 1,000**. It outputs **0.5** and **0.8**.

What are the real fraud probabilities?

Speaker notes

One minute to read, one to think; the odds 1 and 4 are chosen so that the arithmetic is trivial. The punchline of the first answer: a coin-flip score on balanced data means "nothing unusual" on real traffic. Exact values: $1/1000 = 0.100\%$ and $4/1003 = 0.399\%$.



Mini-Exercise

A fraud model trained on a **50/50** sample is used on real traffic, where fraud is **1 in 1,000**. It outputs **0.5** and **0.8**.

What are the real fraud probabilities?

Multiply the odds by $1/999$. Score 0.5: odds 1 $\rightarrow$ $1/999$, about **0.1%** — the base rate itself. Score 0.8: odds 4 $\rightarrow$ $4/999$, about **0.4%**.

Takeaway

A score is a probability when it is **calibrated** — checkable on a reliability diagram, not by accuracy. Maximum likelihood guarantees it only on average and only on the training distribution; **rebalancing** breaks it, and the prior odds repair it. **Regularisation is a prior**, and a penalised fit is a MAP estimate.

Every Loss So Far Is a Likelihood

So far — Lecture 06 meets losses that are not.

you minimised	it is $-\log$ of	the fit is
squared error	Gaussian noise around $\mathbf{w}^\top \mathbf{x}$	the peak (MLE)
cross-entropy	Bernoulli labels with bias $\sigma(\mathbf{w}^\top \mathbf{x})$	the peak (MLE)
$+\lambda \ \mathbf{w}\ ^2$	a Gaussian prior on $\mathbf{w}$	a prior's peak (MAP)
$+\lambda \ \mathbf{w}\ _1$	a Laplace prior on $\mathbf{w}$	a prior's peak (MAP)

Dropped: the spread

how sure we are about $\mathbf{w}$ — **Lecture 07**
keeps it

Dropped: $p(\mathbf{x})$

what a normal input looks like — **Lecture 04**
models it

The lecture in one table; point at the argument map's takeaway box. Read each row as a translation: a loss is a noise model, a penalty is a prior, a fitted model is a peak. Say "so far" out loud: the hinge loss and the 0/1 error are not negative log-likelihoods, and Lecture 06's shortcut (keep the loss) is exactly what remains when there is no probability model behind a loss. The two cards point forward: the villa's band (the spread, Lecture 07) and the unusual transfer ($p(x)$, Lecture 04).

Backup: Classical Regression

For questions: the closed form, R^2 , basis functions, Gauss–Markov and gradient descent
— what students know from earlier courses.

Not presented in the session. Shared units from the Data Science course, included unchanged; Module 2's recap slide points here.



Classification versus Regression

Classification

Y is a finite set of labels.

Will this customer default — yes or no?

Regression

$Y = \mathbb{R}$, the target is a quantity.

What credit line should this customer receive?

The learning machinery is the same; only the shape of Y — and therefore the loss — changes.

Keep the same running example across both to make the point that the task, not the data, decides which you are doing.



The Model

Linear regression predicts the target as a weighted sum of the features:

$$h_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1x_1 + \dots + w_px_p = \mathbf{w}^\top \mathbf{x}$$

with $x_0 = 1$ absorbing the intercept.

The hypothesis space is $\mathbb{R}^{p+1}$: one hypothesis per weight vector.

Name it as a hypothesis space — here a continuous space of finite dimension, which is why calculus rather than enumeration finds the optimum.

The Objective: Residual Sum of Squares

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w} \in \mathbb{R}^{p+1}} \text{RSS}(\mathbf{w}), \quad \text{RSS}(\mathbf{w}) = \sum_{i=1}^n (y_i - \mathbf{w}^\top \mathbf{x}_i)^2$$

RSS measures **effectiveness** — how good the predictions are. Do not confuse it with **efficiency**, which is about runtime and computational effort.

The effectiveness/efficiency distinction is from the source and worth keeping; students routinely say "efficient model" when they mean "accurate model".



One Feature: The Closed Form

For a single feature the minimiser can be written down directly:

Slope

$$\hat{w}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}$$

The covariance of x and y over the variance of x .

Intercept

$$\hat{w}_0 = \bar{y} - \hat{w}_1 \bar{x}$$

The fitted line passes through the point of means.

Both formulas are built from quantities the students defined in the statistics module — covariance and variance. Say that explicitly: regression is not new mathematics, it is the old summary statistics arranged to answer a prediction question.

How Much Variance Is Explained?

$$\text{TSS} = \text{ESS} + \text{RSS}, \quad R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$$

where TSS is the total, ESS the explained and RSS the residual sum of squares (the split holds for a least-squares fit with an intercept).

R^2 is the fraction of the target's variation the model accounts for: 0 means no better than predicting the mean, 1 means an exact fit.

Training R^2 never decreases when a feature is added, however useless the feature. A high training R^2 is not evidence that a model generalises — test R^2 and adjusted R^2 can and do fall.

$R^2 = 0$ is exactly the trivial baseline that always predicts the mean. The warning is the regression counterpart of training error — and the reason a model is judged on data it was not fitted to.

Higher Dimensions

With p features the same objective in matrix form,

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

gives the optimum in one step whenever $\mathbf{X}^\top \mathbf{X}$ is invertible.

Invertibility fails when features are linearly dependent or when $p > n$ — more unknowns than equations. Both are common in practice; regularisation is the standard remedy.

Do not derive the normal equations; state them and spend the time on when they break. The $p > n$ case is the modern one — genomics, text — and motivates regularisation more than the algebra does.

Takeaway

Linear regression fits a weighted sum by minimising squared residuals. Its solution is built from **covariance over variance**, and R^2 reports how much variation it explains — not whether it will generalise.

Linear in the Parameters, Not in the Features

A model is **linear** when it is linear in the **weights** $\mathbf{w}$. The features may be arbitrary transformations of the input.

Plain

$$h(\mathbf{x}) = w_0 + w_1x$$

a straight line

Polynomial

$$h(\mathbf{x}) = w_0 + w_1x + w_2x^2$$

a curve — still a *linear model*

Basis functions

$$h(\mathbf{x}) = \sum_j w_j \phi_j(\mathbf{x})$$

any fixed feature map ϕ

What *linear* forbids: a transformation that has to be learned from the data. The feature map is fixed before fitting; only the weights are learned.

Speaker notes

This is the slide that unlocks the rest: the same least-squares machinery fits curves, interactions and periodic patterns, as long as the transformation is chosen in advance. Ask what is *not* expressible before revealing the note — anything where the transformation itself must be learned, which is what neural networks do.



Consequences

The good

The closed-form solution and all statistical guarantees carry over unchanged.

The catch

The feature map is a **modelling decision**. Choose too rich a basis and the model overfits; too poor and it underfits.

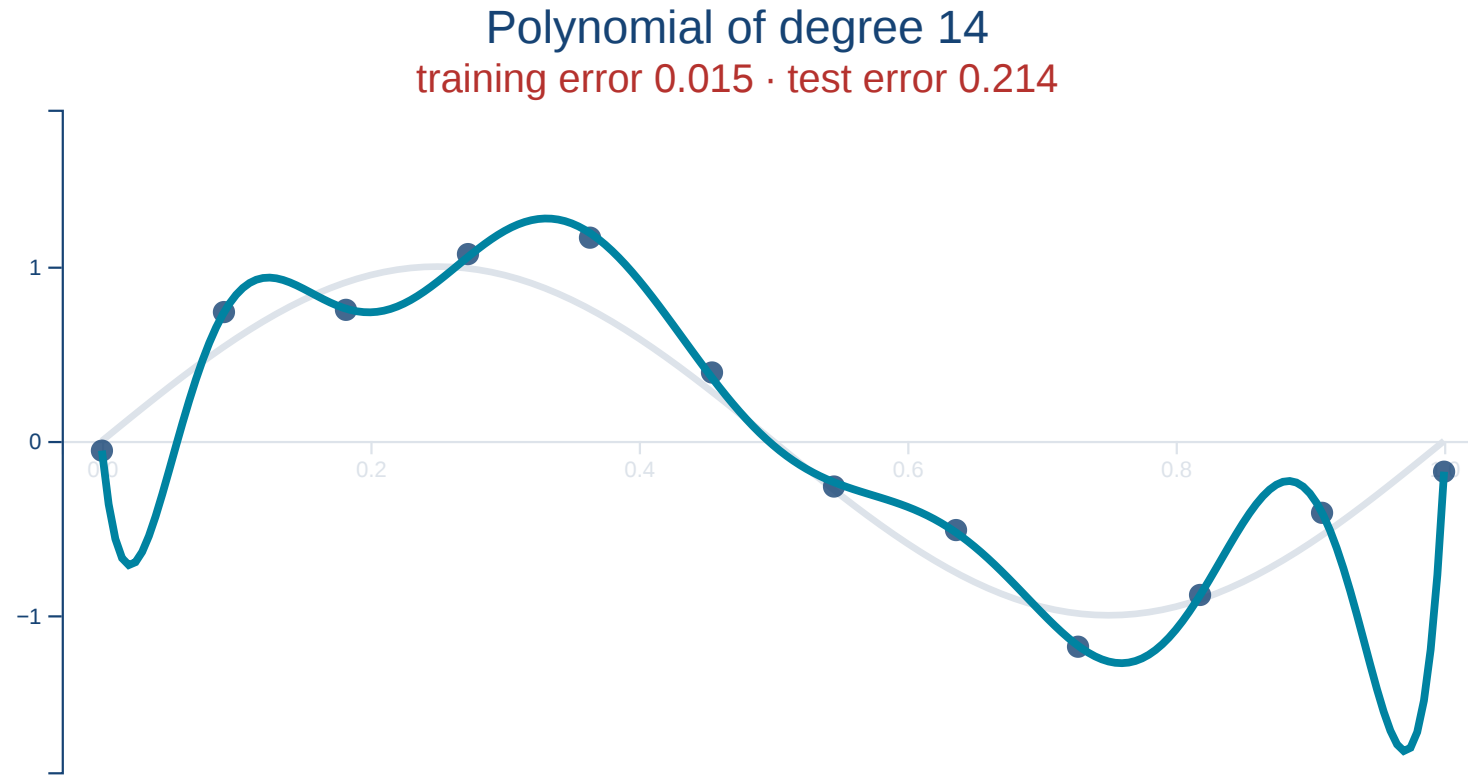
Increasing the polynomial degree turns a dial on the bias–variance axis.

The basis is the inductive bias made explicit. This is the cleanest example in the course of choosing a hypothesis space deliberately.



Turning the Capacity Dial

Linear in the parameters, any basis in the features: a polynomial of degree k is still "linear regression" — and k is a dial.



Speaker notes

Backup, for questions about basis functions. Each click raises the degree: training error falls monotonically, test error bottoms out and climbs again. In today's language: every degree is fitted by keeping the peak; the penalty of Module 4 (a prior) is the other way to turn the dial down.



The Question Nobody Asks

We minimised the *squared* residuals. Why squares — why not absolute deviations, or fourth powers?

Guiding question. Most students have never been given a reason beyond differentiability. There is a real one, and it is worth the slide.



Geometric View

The predictions $\mathbf{X}\mathbf{w}$ live in the **column space** of $\mathbf{X}$ — everything the features can express. The observed $\mathbf{y}$ generally does not.

Least squares picks the point of that space **closest to $\mathbf{y}$ in Euclidean distance**: the orthogonal projection. The residual is perpendicular to every feature.

Consequence

The residual carries no remaining linear information about the features.

The orthogonality condition is the geometric content of the normal equations. It also explains why adding a feature can never increase RSS: the space you project onto only grew.

The Gauss–Markov Theorem

Gauss–Markov (informal). Among all estimators that are

- **linear** in the observed outputs $(y_1, \dots, y_n)$, and
- **unbiased**,

the least-squares estimator has the **lowest variance**. It is the *Best Linear Unbiased Estimator* — BLUE.

Best

lowest variance among the candidates

Linear

a linear function of the observations

Unbiased

right on average over repeated samples

Speaker notes

Spell out what the theorem does *not* say: nothing about non-linear or biased estimators — and biased estimators (ridge, lasso) can beat OLS in prediction error. That is not a contradiction, it is a different criterion.



What This Buys, and What It Costs

The payoff

Minimising RSS yields optimal statistical properties under **mild assumptions** — no normality needed, only a correctly specified linear model, linearly independent features, and zero-mean, uncorrelated errors of constant variance.

The price

Alternatives — absolute error, Huber loss — are **more robust to outliers** but give up the BLUE guarantee.

Squared error is not an arbitrary convenience: it is the loss whose minimiser is the best linear unbiased estimator. Departing from it is legitimate — but it is a trade, and you should be able to name what you traded.

Close the loop with the robustness discussion from descriptive statistics: mean versus median reappears here as squared versus absolute loss. Same trade-off, new context — exactly the spiral the pedagogic concept asks for.

Two Ways to the Same Minimum

Direct (closed form)

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Exact, one step. Requires forming and inverting a $(p+1) \times (p+1)$ matrix — costly for large p , impossible when it is singular.

Iterative (gradient descent)

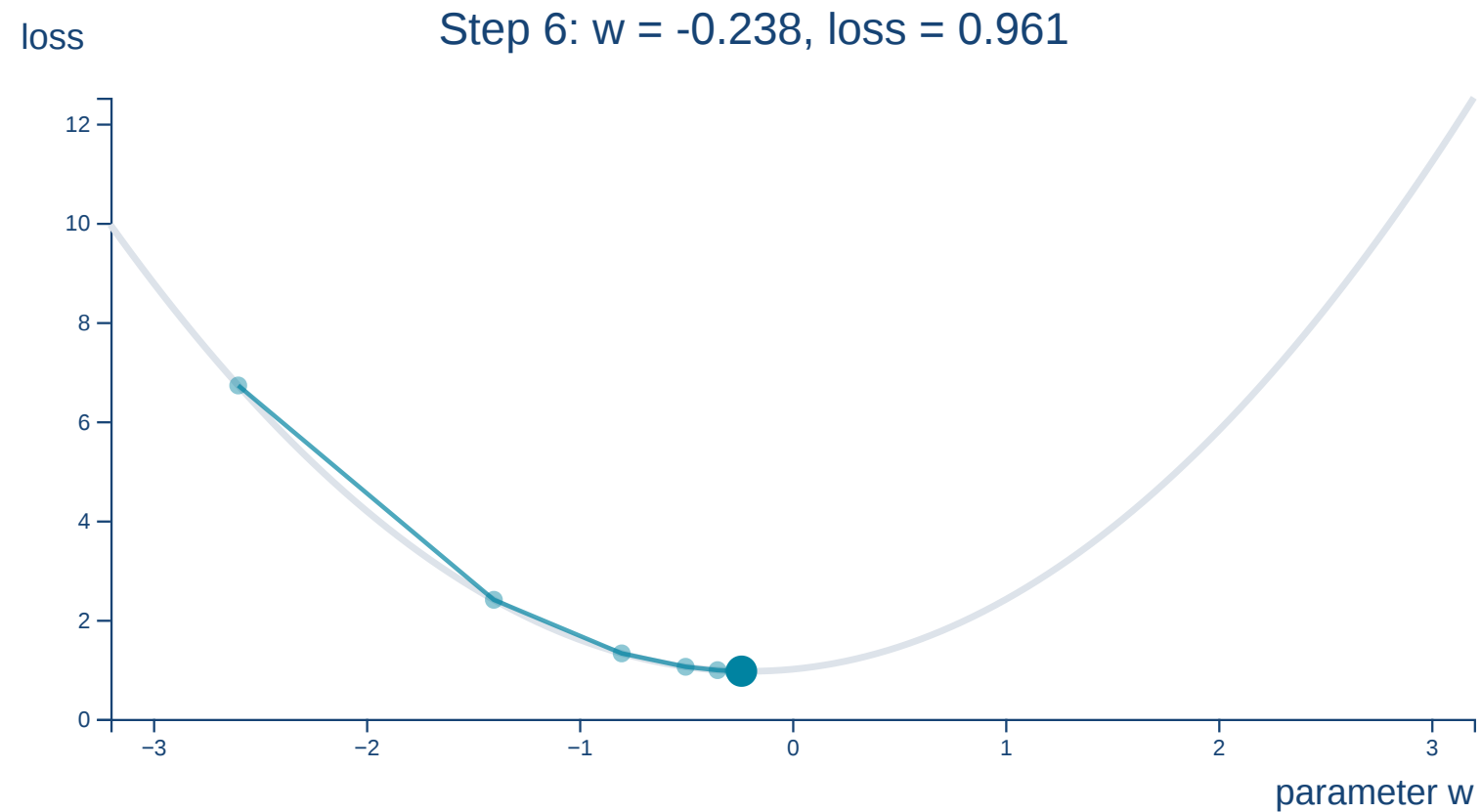
$$\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla \text{RSS}(\mathbf{w})$$

Approximate, many steps. Scales to large p and n , works when the direct solution does not exist, and extends to models with no closed form at all.

The iterative rule is exactly the **LMS algorithm**: its update $w_i \leftarrow w_i + \eta e x_i$ is a gradient step on RSS, taken per example instead of per dataset (the factor 2 is absorbed into η).

The right-hand column is the one that generalises: logistic regression and neural networks have no closed form, so the iterative method is the one that generalises. Make the LMS identification explicit — students otherwise carry two unconnected procedures.

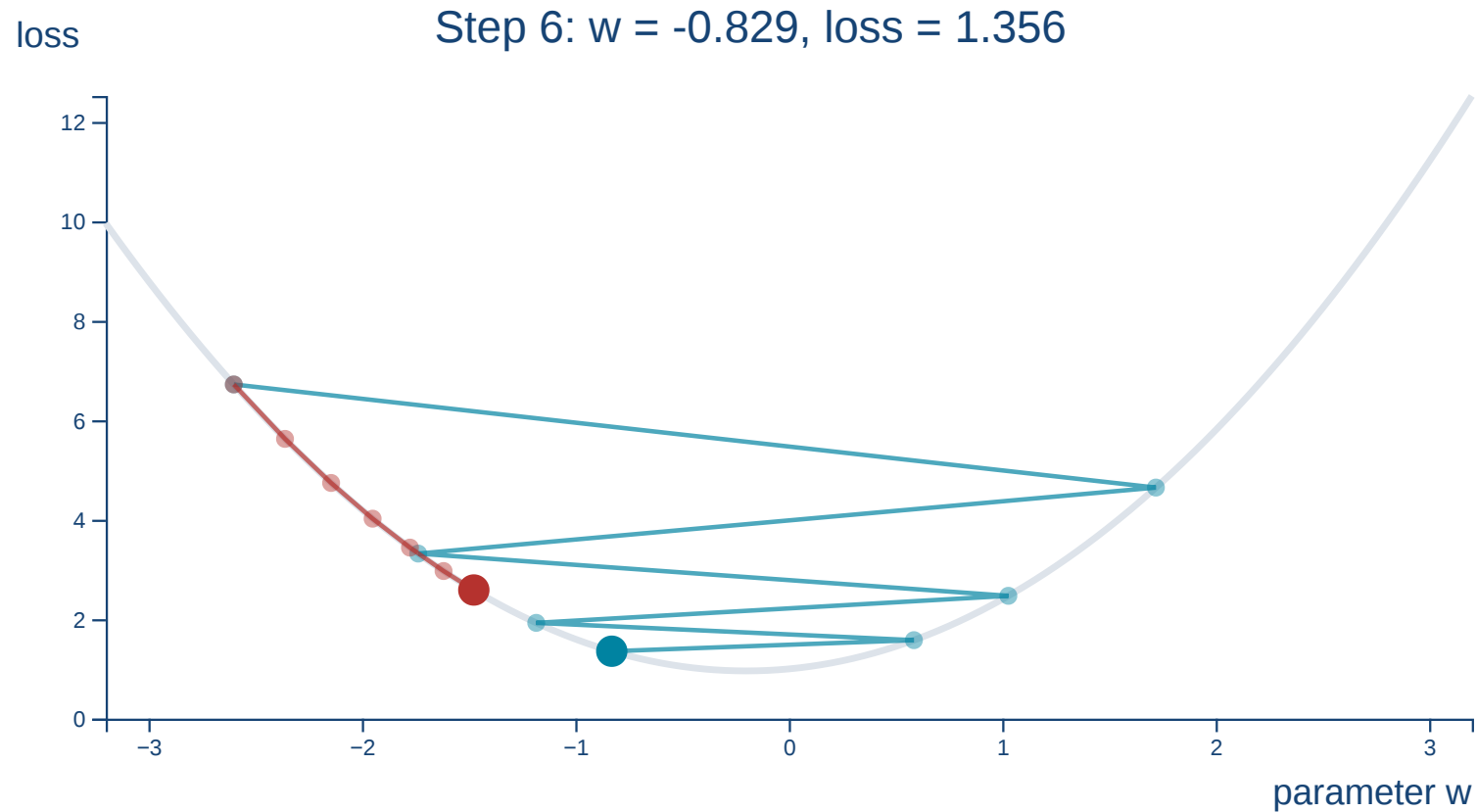
Watching Gradient Descent Work



Stepped viz (P6). Each click is one update. Ask before the first click where the steps will be largest — where the slope is steepest, far from the minimum — and why they shrink as the minimum approaches, without η changing.



The Learning Rate Decides Everything



Teal: $\eta = 0.9$ — overshoots and oscillates across the minimum. Red: $\eta = 0.05$ — stable, but slow: after six steps it has covered about half the distance.



The learning rate is a **hyperparameter**: it is chosen, not learned, and both failure modes are silent — one looks unstable, the other looks converged.

Speaker notes

Predict-then-check: ask which curve belongs to which η before revealing. The "looks converged" failure is the dangerous one — a slow run and a finished run produce the same-looking loss curve over a short window.



Takeaway

Use the **closed form** when the matrix is small and well-conditioned; use **gradient descent** when it is not — and for most models beyond linear regression, no closed form exists at all.